

HERA-PdM: When Does Uncertainty-Gated Escalation Help Hierarchical Edge–Digital-Twin Predictive Maintenance? Evidence from Two Bearing Datasets

Pradeep Somasundaram

Abstract—Hierarchical predictive-maintenance (PdM) systems escalate from edge devices to fog/cloud prognostics, usually on anomaly thresholds or uncalibrated confidence. We study HERA-PdM, in which cross-validated conformalized quantile regression (CQR-CV+) gives remaining-useful-life (RUL) intervals, the edge escalates only when its interval does not fix a unique maintenance action, and high-impact actions are SHAP-supported recommendations for human approval. On 17 FEMTO and 15 XJTU-SY run-to-failure bearings (bearing-grouped cross-validation, three seeds, tuned LSTM/GRU/TCN/CNN-LSTM baselines), CQR-CV+ raised coverage to 0.94–0.96 and 0.93–0.99 (nominal 0.90). Versus point estimates with fixed thresholds, calibrated decisions cut critical snapshots left without action from 49.5% to 0.6% (FEMTO) and 34.2% to 9.2% (XJTU-SY), at the cost of more false maintenance. On gradually degrading XJTU-SY bearings, escalating 79.1% of snapshots matched always-cloud safety with less false maintenance (53.2% vs. 69.1%) and payload; on abrupt FEMTO failures it offered no benefit. The decision-sufficiency gate performed no better than an interval-width gate, and all calibrated policies raised maintenance alarms prematurely for most bearings. Uncertainty-gated hierarchies thus pay off only when degradation is observable early.

Index Terms—predictive maintenance, remaining useful life, conformal prediction, edge computing, Digital Twin, human-in-the-loop, Industry 5.0

I. INTRODUCTION

Predictive maintenance (PdM) estimates the remaining useful life (RUL) of machinery from condition data so that interventions are scheduled before failure [1]. Deep sequence models are the prevailing RUL estimators [2], [3], but in a factory three further requirements arise. Vibration is sensed at the machine, so edge–cloud architectures keep routine inference local [4]; a point estimate must be accompanied by calibrated uncertainty [5], [6]; and Industry 5.0 keeps the human in charge [7], since decision-makers often do not adopt system-generated maintenance advice [8].

Recent systems combine these elements with agents and Digital Twins (DTs) [9]–[12]; our audit of 42 papers (Section II) shows that this *combination* is no longer new. What remains open is the *escalation rule*: hierarchical PdM systems escalate on anomaly thresholds [10] or uncalibrated confidence [13], while calibrated conformal deferral has been studied for classification and numerical cascades [14], [15]. We ask *when* conformally calibrated RUL intervals on the edge are

a useful escalation signal, and at what cost in accuracy, safety, alarm timing, communication and latency.

Contributions:

- 1) **HERA-PdM**, a reproducible four-layer framework with a *decision-sufficiency* gate that escalates only when the edge RUL interval straddles a maintenance-decision boundary, bounding locally resolved unsafe decisions by the edge miscoverage level, and CQR-CV+ calibration on all non-test assets.
- 2) A two-dataset, bearing-grouped benchmark (FEMTO with abrupt, XJTU-SY with gradual degradation) of ten tuned quantile models with and without DT context, showing that CQR-CV+ restores nominal coverage and that larger fog/cloud models help only on the gradual dataset.
- 3) Snapshot-level, per-bearing and sensitivity evidence that calibration removes most unsafe point-threshold decisions, that uncertainty-gated escalation improves on always-cloud inference only under gradual degradation, and that the specific gate matters less than the interval width.

II. RELATED WORK AND RESEARCH GAP

RUL prediction. LSTM [2], multiscale CNN for bearings [3] and TCN [16] models are standard; reviews [1] treat health-stage division, i.e. detecting when degradation becomes observable, as a step preceding RUL prediction, which our two datasets [17], [18] contrast sharply. *DTs and agents.* DTs provide a synchronized virtual asset state [9]; agentic and human-centred DT architectures for PdM have appeared [11], [12], and [12] lists hierarchical DT orchestration with residual-life estimation and explainability as an open question. *Hierarchical PdM.* Ren et al. [4] run a lightweight TCN at the edge and refine RUL in the cloud; HAMA [10] escalates from a z-score edge pre-filter to fog ensembles with SHAP and generated operator text; EC-Distill-ZeroDiag [13] offloads uncertain fault-diagnosis cases from an edge agent to a cloud model. *Uncertainty.* Bayesian, dropout and ensemble RUL uncertainty has been benchmarked in [5]; split conformal and CQR give finite-sample coverage [19], [20], CV+ extends it to cross-fitting [21], and conformal RUL intervals exist for C-MAPSS [6], [22] and bearings [23]. Probabilistic RUL also

TABLE I
CLOSEST RELATED WORK (CODED FROM FULL TEXT, ABSTRACTS OR
PUBLIC CODE). ✓: PRESENT, ○: PARTIAL, -: ABSENT.

Work	DT	Edge	Agents	RUL	Calib.	UQ	Routing	XAI	HIL
Ren et al. [4]	-	✓	-	✓	-	-	-	-	-
HAMA [10]	-	✓	✓	○	-	○ anomaly	-	✓	○
EC-Distill [13]	-	✓	✓	-	-	✓ conf.	✓	○	-
CAB cascade [14]	-	✓	-	-	✓	-	✓	-	-
Wang et al. [23]	-	-	-	✓	✓	-	-	-	-
Xu & Zhang [24]	-	-	○	✓	○	-	-	-	-
HERA-PdM	○	✓	✓	✓	✓	-	✓	✓	✓

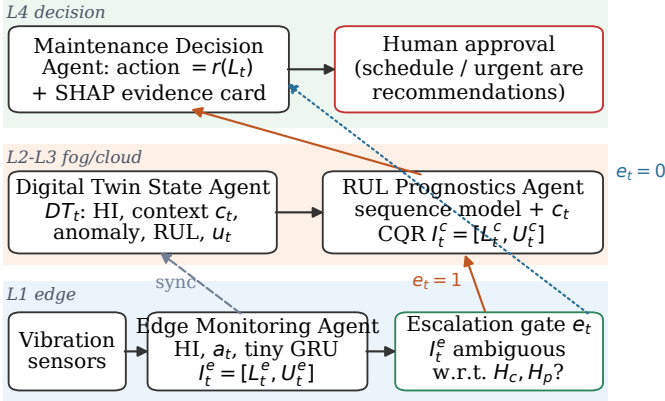


Fig. 1. HERA-PdM. The edge agent computes a calibrated RUL interval I_t^e ; the gate escalates only if I_t^e is not decision-sufficient; the fog/cloud agent uses the DT context c_t ; schedule/urgent actions need human approval.

drives risk-aware maintenance policies [24], and confidence-based [25] and conformal [14], [15] deferral are the closest methodological precedents.

Gap. To the best of our review (Table I), no study uses conformal RUL intervals computed on the edge as the escalation rule of a hierarchical PdM system, relates escalation to maintenance-decision boundaries, and evaluates it against the gates of existing systems on public run-to-failure data with different degradation regimes. The contribution is methodological and evaluative; the architecture is not new.

III. HERA-PdM METHODOLOGY

Fig. 1 shows the four layers. “Agents” are software components with fixed responsibilities and messages; no language model is used anywhere.

Edge Monitoring Agent. Each vibration snapshot is reduced to $x_t \in \mathbb{R}^{28}$ (per axis: RMS, kurtosis, skewness, peak-to-peak, crest, impulse and shape factors, spectral centroid, six log band energies), z-scored against the asset’s first snapshots and compressed by $\text{sign}(z) \log(1+|z|)$. The health index is $h_t = \log(1 + \frac{1}{d} \sum_j |z_{t,j}|)$; the anomaly score a_t is the smoothed h_t standardized by the asset’s raw baseline, with alarm threshold τ_a at the 99th percentile of healthy training snapshots. A tiny model maps a short window to three RUL quantiles.

Digital Twin State Agent. The twin is a synchronized state record $DT_t = \{\text{condition, sensor state, } h_t, \text{ context } c_t, \hat{R}_t,$

$I_t, u_t, \text{ anomaly state}\}$ updated from a 12-byte sync (t, h_t, a_t) per snapshot; $c_t \in \mathbb{R}^6$ holds fast/slow averages of h_t , their difference, running maximum, slope and cumulative excess over baseline. Absolute operating time is excluded because lifetimes vary by an order of magnitude.

RUL Prognostics Agent. A sequence model g_ϕ maps a longer window X_t and c_t to non-crossing quantiles $(\hat{q}_t^{\alpha/2}, \hat{q}_t^{0.5}, \hat{q}_t^{1-\alpha/2})$ trained with the pinball loss on $y_t = \min(\text{RUL}_t, R_{\max})$.

Calibration (CQR-CV+). The non-test assets are split into $K=4$ condition-balanced groups; inner model $g^{(-k)}$ is trained without group k , which yields out-of-fold scores $s_i = \max(\hat{q}_i^{\alpha/2} - y_i, y_i - \hat{q}_i^{1-\alpha/2})$ for every non-test snapshot i in group $k(i)$. Following CV+ [21] with CQR scores [19],

$$L_t = Q_\alpha^- \{\hat{q}_t^{\alpha/2, (-k(i))} - s_i\}, \quad (1)$$

$$U_t = Q_{1-\alpha}^+ \{\hat{q}_t^{1-\alpha/2, (-k(i))} + s_i\}, \quad (2)$$

where Q_α^- and $Q_{1-\alpha}^+$ are the $\lfloor \alpha(n+1) \rfloor$ -th and $\lceil (1-\alpha)(n+1) \rceil$ -th smallest values over the n calibration snapshots; $\hat{R}_t$ is the mean inner median. CV+ guarantees coverage $\geq 1 - 2\alpha$ (typically $\approx 1 - \alpha$) if assets are exchangeable; snapshots within an asset are dependent, so the guarantee is approximate and we test it. Split CQR on three held-out bearings is the baseline calibrator.

Decision and escalation. With critical and planning horizons H_c and H_p , the action of a value v is $r(v) = \text{URGENT}$ if $v \leq H_c$, SCHEDULE if $H_c < v \leq H_p$, else CONTINUE; the Maintenance Decision Agent acts on $\pi_t = r(L_t)$ (CONTINUE becomes INSPECT if $a_t > \tau_a$). SCHEDULE/URGENT are recommendations with an explanation card (RUL, interval, anomaly evidence, top-3 SHAP [26] contributors, rationale) that require approval. With $B = \{H_c, H_p\}$, the edge escalates if its interval is ambiguous or contradicts anomaly evidence:

$$e_t = \mathbb{I}[\exists h \in B: L_t^e \leq h < U_t^e] \vee \mathbb{I}[a_t > \tau_a \wedge r(L_t^e) = C], \quad (3)$$

where C denotes CONTINUE. A local CONTINUE needs $L_t^e > H_p \geq H_c$, so a critical asset ($y_t \leq H_c$) is left running locally only if $y_t \notin I_t^e$:

$$\Pr(e_t=0, \pi_t=\text{CONTINUE}, y_t \leq H_c) \leq \Pr(y_t \notin I_t^e), \quad (4)$$

and false local URGENT actions are bounded likewise. Unlike a width threshold, the gate ignores wide intervals inside one action region and escalates narrow ones crossing a boundary.

Cost model. Payloads are counted in float32: always-cloud streams x_t (112 B/snapshot); HERA sends the 12 B sync plus the untransmitted part of the window when $e_t=1$. Latency is edge time $+ e_t(\text{RTT} + \text{cloud time})$ with an assumed RTT of 50 ms.

IV. EXPERIMENTAL SETUP

Data. FEMTO-ST/PRONOSTIA [17]: 17 bearings under three conditions, 0.1 s snapshots every 10 s (24,889 snapshots), full test trajectories; several bearings fail abruptly. XJTU-SY [18]: 15 bearings under 35 Hz/12 kN, 37.5 Hz/11 kN and 40 Hz/10 kN, 1.28 s snapshots every

TABLE II

RUL ACCURACY AND CALIBRATION ON HELD-OUT BEARINGS (MEAN $\pm$ STD OVER 3 SEEDS; NOMINAL COVERAGE 0.90). RAW: UNCALIBRATED QUANTILES; SPLIT: CQR CALIBRATED ON 3 HELD-OUT BEARINGS PER FOLD; CV+: CQR-CV+ CALIBRATED ON ALL NON-TEST BEARINGS. MARKS: ARCHITECTURE SELECTED ON INNER-VALIDATION PINBALL LOSS FOR FEMTO ($\dagger$) AND XJTU-SY (*).

Model	FEMTO/PRONOSTIA (17 bearings, $R_{\max}=3000$ s)					XJTU-SY (15 bearings, $R_{\max}=6000$ s)				
	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)
Edge-CNN* (split)	1114 $\pm$ 40	851 $\pm$ 19	0.37	0.88 $\pm$ 0.02	2080	2278 $\pm$ 389	1859 $\pm$ 330	0.65	0.97 $\pm$ 0.00	3716
Edge-GRU $\dagger$ (split)	820 $\pm$ 34	518 $\pm$ 62	0.71	0.84 $\pm$ 0.04	1877	2531 $\pm$ 642	2004 $\pm$ 601	0.85	0.87 $\pm$ 0.08	4184
LSTM (split)	1208 $\pm$ 118	772 $\pm$ 102	0.50	0.85 $\pm$ 0.04	2263	1516 $\pm$ 127	886 $\pm$ 104	0.90	0.91 $\pm$ 0.01	2436
LSTM+DT $\dagger$ (split)	1124 $\pm$ 48	751 $\pm$ 54	0.37	0.84 $\pm$ 0.04	2154	1698 $\pm$ 58	1130 $\pm$ 68	0.89	0.88 $\pm$ 0.03	3107
GRU (split)	1171 $\pm$ 114	751 $\pm$ 84	0.40	0.86 $\pm$ 0.03	2151	1516 $\pm$ 39	1027 $\pm$ 73	0.89	0.89 $\pm$ 0.02	2404
GRU+DT* (split)	1140 $\pm$ 89	771 $\pm$ 66	0.32	0.89 $\pm$ 0.03	2286	1647 $\pm$ 94	1226 $\pm$ 153	0.86	0.83 $\pm$ 0.06	3042
TCN (split)	1159 $\pm$ 70	827 $\pm$ 72	0.16	0.85 $\pm$ 0.02	2241	1666 $\pm$ 119	1303 $\pm$ 145	0.52	0.95 $\pm$ 0.01	3691
TCN+DT (split)	1153 $\pm$ 136	855 $\pm$ 115	0.16	0.89 $\pm$ 0.01	2456	1496 $\pm$ 118	1100 $\pm$ 138	0.62	0.81 $\pm$ 0.11	3477
CNN-LSTM (split)	1106 $\pm$ 74	674 $\pm$ 34	0.40	0.85 $\pm$ 0.02	2119	1335 $\pm$ 3	772 $\pm$ 21	0.85	0.90 $\pm$ 0.01	2416
CNN-LSTM+DT (split)	1108 $\pm$ 39	747 $\pm$ 54	0.28	0.89 $\pm$ 0.01	2355	1446 $\pm$ 30	895 $\pm$ 33	0.84	0.91 $\pm$ 0.00	2686
LSTM+DT (MSE point)	1090 $\pm$ 161	780 $\pm$ 129	–	–	–	1656 $\pm$ 41	1134 $\pm$ 54	–	–	–
GRU+DT (MSE point)	1166 $\pm$ 121	858 $\pm$ 92	–	–	–	1689 $\pm$ 115	1293 $\pm$ 159	–	–	–
TCN+DT (MSE point)	1232 $\pm$ 110	925 $\pm$ 89	–	–	–	1539 $\pm$ 95	1156 $\pm$ 107	–	–	–
CNN-LSTM+DT (MSE point)	1135 $\pm$ 45	825 $\pm$ 37	–	–	–	1468 $\pm$ 29	897 $\pm$ 14	–	–	–
Selected edge (CV+)	753 $\pm$ 5	501 $\pm$ 22	0.75	0.94 $\pm$ 0.01	2193	1938 $\pm$ 251	1677 $\pm$ 232	0.69	0.99 $\pm$ 0.01	4134
Selected cloud+DT (CV+)	892 $\pm$ 27	665 $\pm$ 14	0.28	0.96 $\pm$ 0.02	2506	1422 $\pm$ 38	1077 $\pm$ 59	0.91	0.93 $\pm$ 0.01	3030
Selected cloud, no DT (CV+)	954 $\pm$ 16	653 $\pm$ 13	0.35	0.95 $\pm$ 0.02	2551	1305 $\pm$ 54	757 $\pm$ 66	0.90	0.94 $\pm$ 0.01	2337

TABLE III

HIERARCHICAL POLICIES, ABLATIONS AND SYSTEM COST (MEAN $\pm$ STD OVER 3 SEEDS, ALL HELD-OUT BEARINGS). ESC.: ESCALATED SNAPSHOTS; UNSAFE: CRITICAL SNAPSHOTS ($y \leq H_c$) LEFT WITHOUT A MAINTENANCE ACTION; FM: FALSE MAINTENANCE ON HEALTHY SNAPSHOTS ($y > H_p$); B/s: PAYLOAD BYTES PER SNAPSHOT (ANALYTIC). $\dagger$ ESCALATION BUDGET MATCHED TO HERA ON CALIBRATION DATA.

Policy	FEMTO/PRONOSTIA					XJTU-SY				
	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s
Always-edge	0.0	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	752	0	0.0	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1485	0
Always-cloud	100.0	0.0 $\pm$ 0.0	96.9 $\pm$ 2.7	891	112	100.0	9.2 $\pm$ 1.7	69.1 $\pm$ 6.1	1421	112
Cloud point + fixed thr.	100.0	49.5 $\pm$ 2.7	21.7 $\pm$ 6.8	1090	112	100.0	34.2 $\pm$ 9.2	8.3 $\pm$ 2.1	1689	112
HERA-full	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA w/o anomaly term	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA-no-DT	97.1	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	953	123	79.1	12.1 $\pm$ 4.3	25.5 $\pm$ 7.7	1303	104
HERA-no-UQ (point margin)	48.1	56.2 $\pm$ 5.3	15.1 $\pm$ 2.8	989	89	67.8	34.2 $\pm$ 9.2	8.0 $\pm$ 2.2	2062	91
HERA-fixed (anomaly trig.)	3.9	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	760	27	5.4	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1541	20
Width gate $\dagger$	97.2	0.6 $\pm$ 1.1	95.1 $\pm$ 0.5	892	123	79.2	9.2 $\pm$ 1.7	53.3 $\pm$ 13.3	1381	104
Linear score gate $\dagger$	94.3	0.6 $\pm$ 1.1	95.7 $\pm$ 0.9	892	121	77.9	9.2 $\pm$ 1.7	53.8 $\pm$ 13.3	1378	103
Random gate $\dagger$	96.4	0.0 $\pm$ 0.0	96.8 $\pm$ 2.6	887	124	71.6	6.5 $\pm$ 1.4	71.4 $\pm$ 6.3	1441	124

minute (9,216 snapshots), mostly gradual degradation. RUL caps are $R_{\max}=3000$ s (FEMTO) and 6000 s (XJTU-SY); $H_c=0.2R_{\max}$, $H_p=0.6R_{\max}$; windows are 16/64 and 8/32 snapshots (edge/cloud).

Protocol. Condition-balanced bearing-grouped folds (4 for FEMTO, 5 for XJTU-SY) test every bearing once. Each architecture’s hyper-parameters (width 32–128, 8 or 25 epochs, input noise) and the edge/cloud architecture choice were selected on inner validation bearings from training bearings only (FEMTO: Edge-GRU/LSTM+DT; XJTU-SY: Edge-CNN/GRU+DT). Three seeds per model; metrics are pooled over test bearings and reported as mean $\pm$ std; paired Wilcoxon tests use bearings as units.

Baselines and metrics. LSTM, GRU, TCN [16] and CNN-LSTM with and without DT context, MSE point models; always-edge, always-cloud and point-with-fixed-threshold

policies; ablations without uncertainty (point margin), DT or anomaly term; anomaly-trigger (as in [10]), width (confidence deferral [25]), linear-score and random gates at HERA’s budget. We report RMSE, MAE, coverage (Cov.), mean interval width (MPIW), *unsafe rate* (critical snapshots with CONTINUE/INSPECT), false maintenance (FM, SCHEDULE/URGENT when $y_t > H_p$), escalation, payload, latency (single-thread host CPU, Apple M2 Pro; no energy measured) and, per bearing, the timing of the first sustained maintenance alarm.

V. RESULTS AND DISCUSSION

RQ1: accuracy. Table II shows opposite regimes. On FEMTO, the 2.5k-parameter edge model was most accurate (RMSE 753 s with CQR-CV+ vs. 892 s for LSTM+DT), because abrupt failures leave larger models nothing to exploit.

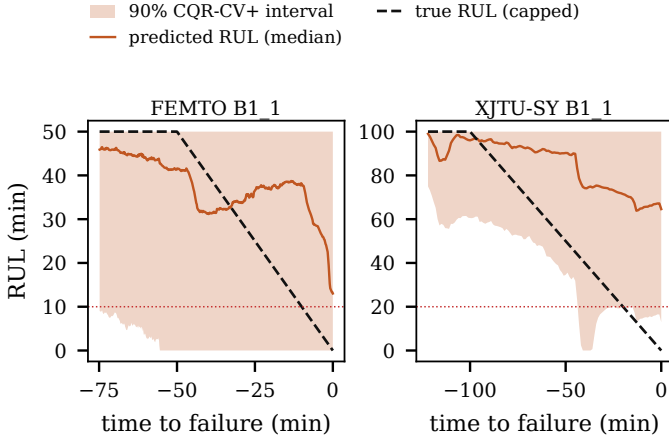


Fig. 2. Fog/cloud median RUL and 90% CQR-CV+ interval before failure (seed 0; per dataset the test bearing with most degradation-phase snapshots). Dotted line: H_c .

TABLE IV

TOP: OUTCOME OF EACH BEARING'S FIRST SUSTAINED (3 SNAPSHOTS) SCHEDULE/URGENT RECOMMENDATION (NUMBER OF BEARINGS, MEAN OVER 3 SEEDS): T TIMELY ($H_c \leq \text{RUL} \leq R_{\max}$), L LATE ($\text{RUL} < H_c$ OR NEVER), P PREMATURE ($\text{RUL} > R_{\max}$); LEAD: MEDIAN RUL AT TIMELY ALARMS (MIN). BOTTOM: SENSITIVITY OF UNSAFE/FM (%) FOR HERA-FULL AND THE POINT POLICY TO THE HORIZONS AND TO $R_{\max}$.

Policy	FEMTO (17 bearings)				XJTU-SY (15 bearings)			
	T	L	P	lead	T	L	P	lead
Always-edge	1.0	0.0	16.0	38	2.0	0.0	13.0	44
Always-cloud	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Point + thr.	1.0	5.3	10.7	40	5.0	4.0	6.0	54
HERA-full	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Width gate	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Setting	HERA		Point		HERA		Point	
$H_c, H_p=0.1, 0.4R_{\max}$	13/68		54/15		16/15		50/5	
$H_c, H_p=0.2, 0.6R_{\max}$	1/95		49/22		9/53		34/8	
$H_c, H_p=0.3, 0.8R_{\max}$	0/100		37/39		2/81		23/42	
$R_{\max} \times \frac{1}{2}$ (seed 0)	24/24		31/23		6/21		54/5	
$R_{\max} \times 2$ (seed 0)	0/100		19/39		0/77		21/31	

On XJTU-SY, fog/cloud models were clearly better (best split-trained 1335 s, CNN-LSTM) than the selected edge model (2278 s). DT context lowered RMSE for 3 of 4 architectures on FEMTO but only 1 on XJTU-SY; cross-fitted ensembles (CV+) also improved point accuracy.

RQ2: calibration. Split CQR with three calibration bearings under-covered on FEMTO (0.84–0.89). CQR-CV+, which calibrates on all non-test bearings, reached 0.94 (edge) and 0.96 (cloud) on FEMTO and 0.93–0.99 on XJTU-SY, at the price of wider intervals (MPIW 2193 s and 3030 s). Per-bearing coverage still varied strongly (lowest cloud coverage 0.30 and 0.07), so exchangeability between bearings remains the binding assumption. Fig. 2 shows intervals narrowing once degradation is visible, far earlier on XJTU-SY.

RQ4: unsafe decisions. Point estimates with fixed thresholds left 49.5% (FEMTO) and 34.2% (XJTU-SY) of critical snapshots without a maintenance action, versus 0.6% and 9.2% for HERA (Table III); across bearings the reductions were

significant ($p < 0.001$, $p = 0.004$), and so were the increases in false maintenance ($p < 0.001$ on both). Sweeping α or a point margin δ (Fig. 3a,c) shows that calibrated and margin-adjusted point policies lie close to one frontier: calibration contributes a principled, held-out choice of operating point rather than better attainable decisions.

RQ3 and RQ5: escalation and system trade-off. On FEMTO, 90% edge intervals (2193 s on a 3000 s range) almost always straddled H_p , so HERA escalated 97.1% of snapshots, sent more payload than always-cloud streaming (123.0 vs. 112.0 B) and gave no decision benefit. On XJTU-SY, HERA escalated 79.1%, matched always-cloud safety (9.2% vs. 9.2%) with lower false maintenance (53.2% vs. 69.1%), better RMSE (1381 vs. 1421 s), 104.5 instead of 112.0 B/snapshot and mean latency 40.1 vs. 50.6 ms. Two caveats temper this: per bearing, the FM difference to always-cloud was non-zero for only 2 bearings (the gain comes from the two long-lived condition-3 bearings), and the edge-only calibrated policy traced a better frontier than HERA (Fig. 3c) because the conservative edge intervals rarely under-predict. At matched budget, the width gate gave identical results (9.2%/53.3%) and the linear-score gate nearly so (Fig. 3b,d); the anomaly trigger escalated only 5.4% (anomaly AUROC 0.81/0.80). Relaxing α_e to 0.3 reduced escalation to 83.0%/67.9% while unsafe decisions rose to 7.7%/9.2%. Removing the DT context lowered FM on XJTU-SY (25.5%) at higher unsafe rate (12.1%), so the DT context is not uniformly beneficial.

Per-bearing timing and sensitivity. Table IV exposes the main practical limitation: because the 90% lower bound falls below H_p early, every calibrated policy raised its first sustained maintenance alarm while the true RUL still exceeded $R_{\max}$ for 16.0 of 17 FEMTO and 13.0 of 15 XJTU-SY bearings, whereas the point policy was late for 5.3 and 4.0 bearings. Neither is acceptable for scheduling; per-snapshot marginal coverage is the wrong target for alarm timing. In all five horizon settings and all four halved/doubled- $R_{\max}$ runs, HERA left fewer critical snapshots without action than the point policy; with a halved cap on XJTU-SY, HERA reached low false maintenance at the always-cloud unsafe rate (Table IV).

Explainability and oversight. Expected-gradient SHAP on 300 degradation-phase snapshots ranked low-frequency band energies and the spectral centroid (FEMTO) and the DT fast health-index average and mid- and high-frequency band energies (XJTU-SY) highest; additivity errors were 16% and 3% of the explained deviation. An automatically generated card (FEMTO) reads:

Asset Bearing1_3, snapshot 2078. **RUL** 578 s, 90% interval [0, 2138] s. **Anomaly** $a_t=15.8$ (threshold 17.04, no alarm). **Top SHAP** h_band0_1k (-1489 s), v_band2_4k (-1244 s), $h_centroid$ (+1175 s). **Recommendation** urgent maintenance — requires human approval. **Rationale** lower bound $0s \leq H_c$.

All fields come from the predictors via a fixed template. Approval requests covered 95.5% (FEMTO) and 55.8% (XJTU-SY) of HERA snapshots, an operator load that shows calibrated but wide intervals shift burden to humans.

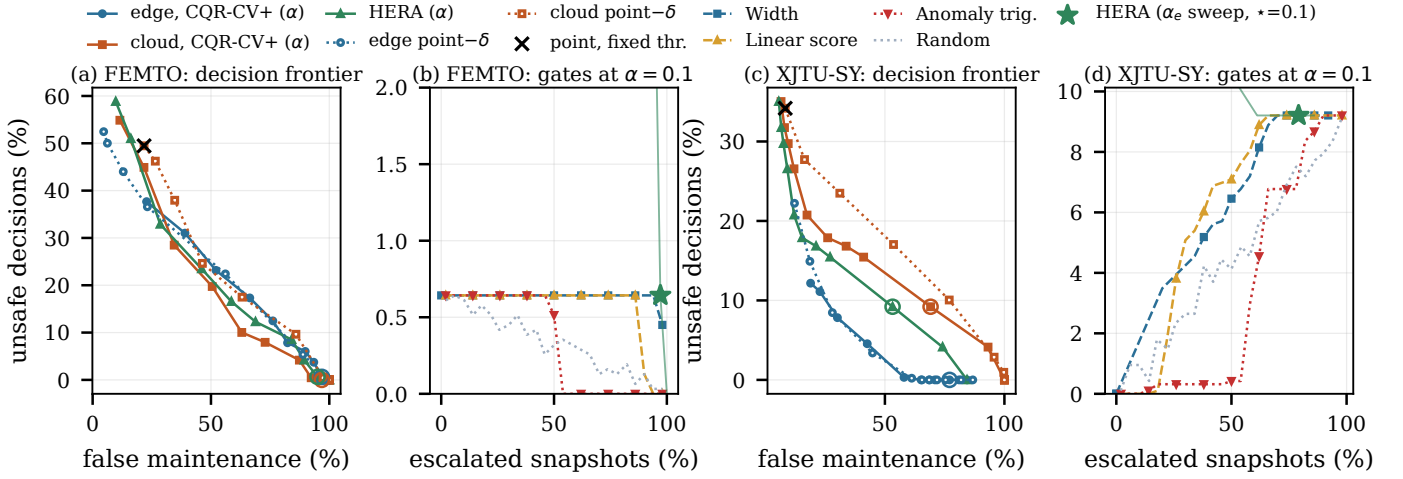


Fig. 3. (a,c) Unsafe decisions vs. false maintenance when sweeping α (calibrated; circles at $\alpha=0.1$) or a margin δ (point). (b,d) Unsafe decisions vs. escalation budget for gates on the same $\alpha=0.1$ edge intervals; green line: HERA with α_e swept.

Limitations. Two bearing datasets (32 bearings); approximate exchangeability; a lightweight state-based DT; host-CPU latency and analytic communication; $R_{\max}$ sensitivity with one seed; no deployment and no user study of the explanation cards.

VI. CONCLUSION

Conformal RUL intervals from cross-fitted calibration removed most unsafe decisions of point-threshold maintenance on two bearing datasets. Using them to gate edge-to-cloud escalation paid off only on gradually degrading bearings, where it matched always-cloud safety with less false maintenance and communication; on abruptly failing bearings it escalated almost everything. The specific decision-sufficiency rule was no better than a width gate, and per-snapshot calibration produced premature bearing-level alarms. Future hierarchical PdM should calibrate alarm timing rather than snapshot coverage and should decide escalation from observed degradation onset.

REFERENCES

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: A systematic review from data acquisition to RUL prediction," *Mechanical Systems and Signal Processing*, vol. 104, pp. 799–834, 2018.
- [2] S. Zheng, K. Ristovski, A. Farahat, and C. Gupta, "Long short-term memory network for remaining useful life estimation," in *Proc. IEEE Int. Conf. Prognostics and Health Management (ICPHM)*, 2017, pp. 88–95.
- [3] J. Zhu, N. Chen, and W. Peng, "Estimation of bearing remaining useful life based on multiscale convolutional neural network," *IEEE Transactions on Industrial Electronics*, vol. 66, pp. 3208–3216, 2019.
- [4] L. Ren, Y. Liu, X. Wang, J. Lü, and M. J. Deen, "Cloud-edge-based lightweight temporal convolutional networks for remaining useful life prediction in IIoT," *IEEE Internet of Things Journal*, vol. 8, no. 16, pp. 12 578–12 587, 2021.
- [5] L. Basora, A. Viens, M. Arias Chao, and X. Olive, "A benchmark on uncertainty quantification for deep learning prognostics," *Reliability Engineering & System Safety*, vol. 253, p. 110513, 2025.
- [6] A. Javanmardi and E. Hüllermeier, "Conformal prediction intervals for remaining useful lifetime estimation," *International Journal of Prognostics and Health Management*, vol. 14, no. 2, 2023.
- [7] X. Xu, Y. Lu, B. Vogel-Heuser, and L. Wang, "Industry 4.0 and industry 5.0— inception, conception and perception," *Journal of Manufacturing Systems*, vol. 61, pp. 530–535, 2021.
- [8] B. van Oudenhoven, P. Van de Calseyde, R. Basten, and E. Demerouti, "Predictive maintenance for industry 5.0: behavioural inquiries from a work system perspective," *International Journal of Production Research*, vol. 61, pp. 7846–7865, 2023.
- [9] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Transactions on Industrial Informatics*, vol. 15, pp. 2405–2415, 2019.
- [10] R. B. Saleh, K. P. Dinh, B. Villanyi, and T.-S. Hy, "HAMA: A hierarchical adaptive multi-agent architecture for industrial IoT predictive maintenance," *IEEE Access*, vol. 14, pp. 142 595–142 613, 2026.
- [11] L. Mendonça, J. Barbosa, and P. Leitão, "A human-centred architecture integrating digital twin and agentic AI for predictive maintenance," in *Proc. IEEE Int. Conf. Industrial Technology (ICIT)*, 2026, pp. 1–6.
- [12] K. A. Coulibaly and M. Hamlich, "Multi-agent system-driven digital twins for predictive maintenance: architectures, technologies and open research challenges," *arXiv preprint arXiv:2607.21873*, 2026.
- [13] Z. Yao, Y. Xu, T. Xia, Z. Chen, E. Pan, and L. Xi, "EC-Distill-ZeroDiag: a cloud-edge collaborative framework for zero-shot industrial fault diagnosis via large language model distillation," *Journal of Intelligent Manufacturing*, 2026.
- [14] J. Huang, S. Park, N. Paoletti, and O. Simeone, "Reliable inference in edge-cloud model cascades via conformal alignment," *arXiv preprint arXiv:2510.17543*, 2025.
- [15] Q. Hou, S. Park, M. Zecchin, Y. Cai, G. Yu, and O. Simeone, "Online conformal probabilistic numerics via adaptive edge-cloud offloading," *arXiv preprint arXiv:2503.14453*, 2025.
- [16] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.
- [17] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, and C. Varnier, "PRONOSTIA: An experimental platform for bearings accelerated degradation tests," in *Proc. IEEE Int. Conf. Prognostics and Health Management (PHM'12)*, Denver, CO, USA, 2012, pp. 1–8, hAL: hal-00719503.
- [18] B. Wang, Y. Lei, N. Li, and N. Li, "A hybrid prognostics approach for estimating remaining useful life of rolling element bearings," *IEEE Transactions on Reliability*, vol. 69, no. 1, pp. 401–412, 2020.
- [19] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [20] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [21] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Predictive

inference with the jackknife+,” *The Annals of Statistics*, vol. 49, no. 1, 2021.

- [22] A. Moccaldi, C. Conte, R. C. Ghosh, and F. Moscato, “A robust conformal framework for IoT-based predictive maintenance,” *Future Internet*, vol. 17, p. 244, 2025.
- [23] S. Wang, Y. Vidal, and F. Pozo, “Uncertainty-aware bearing remaining useful life prediction based on conformal prediction,” in *Proc. PHM Society European Conference*, vol. 9, no. 1, 2026, pp. 1–10.
- [24] Z. Xu and Q. Zhang, “Predictive maintenance optimization for industrial equipment via reliable prognosis and risk-aware reinforcement learning,” *Complex & Intelligent Systems*, vol. 12, 2025.
- [25] W. Jitkrittum, N. Gupta, A. K. Menon, H. Narasimhan, A. S. Rawat, and S. Kumar, “When does confidence-based cascade deferral suffice?” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [26] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.